

Eightfold Multi-Source Candidate Data Transformer

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Goal: turn messy structured and unstructured candidate inputs into canonical, explainable, deterministic profiles. The system prefers honestly-empty fields over invented values and keeps every chosen or discarded value traceable through provenance.

Pipeline

- Detect inputs from CLI/UI: recruiter CSV rows, GitHub username, and resume PDF. Each adapter returns RawRecord objects with source_status ok, partial, or failed.
- Extract source fields with adapter-specific logic. CSV maps semi-structured recruiter columns; GitHub uses public REST data; PDF uses pdf-parse plus a Flate text-stream fallback for PDFKit-style resumes.
- Normalize emails, phones to E.164, countries to ISO-3166 alpha-2, dates to YYYY-MM or YYYY, and skills to canonical names.
- Group records into candidate clusters using normalized email/phone keys, then name/GitHub fallback only when contact keys are absent.
- Merge each cluster into a CanonicalProfile, validate it, optionally project to runtime output config, and validate the projection before returning JSON.

Canonical Schema and Formats

- candidate_id is cand_ plus a deterministic SHA-256 prefix from sorted normalized emails/phones, with name fallback only if contact keys are absent.
- Profile fields: full_name, emails, phones, location, links, headline, years_experience, skills, experience, education, provenance, overall_confidence.
- Phones are E.164; countries are ISO-3166 alpha-2; dates are YYYY-MM where month is known; skills are lowercase canonical labels such as javascript, reactjs, and machine-learning.

Merge, Conflict Resolution, and Confidence

- Source trust is deterministic and based on source type plus data richness. Higher-trust scalar values win; losing conflicting values are retained in provenance as discarded_conflict.
- Emails, phones, skills, education, and experience are unioned/deduplicated. Experience keeps the richer duplicate when CSV and PDF describe the same role.
- Confidence is a pure function of available evidence, agreement, and source trust. Profiles without any contact key are capped at 0.40 to avoid wrong-but-confident output.

Runtime Projection Config

- Projection is separate from merge. Config fields select/rename canonical paths, apply requested normalization, include confidence, and choose missing handling: null, omit, or error.
- The projection layer supports indexed paths such as emails[0] and array paths such as skills[].name, then validates the projected schema with zod.

Edge Cases and Scope

- Malformed source: adapter returns failed or partial and the pipeline continues if at least one valid source remains.
- Multi-row CSV: distinct candidates are not merged; each contact cluster becomes its own profile.
- Unparseable resume PDF: fallback decoder recovers text streams when the primary parser fails.
- Missing contact info: candidate_id falls back to normalized name and confidence is capped.
- Descoped under time pressure: deep NLP entity resolution, LinkedIn scraping/auth, ATS JSON variants, and smart GitHub retry-after scheduling.